



Original article

Major adverse cardiovascular events among patients with type 2 diabetes mellitus: A prospective study



David Martín-Enguix^a, Juan Carlos Aguirre Rodríguez^a, Abraham Hidalgo Rodríguez^b,
María Guisasola Cárdenas^a, María Sánchez Cambroneró^a, María Nieves Generoso Torres^a,
Manuel Joaquín Castillo^{c,*}, Francisco José Amaro-Gahete^{c,d,e,*}

^a Centro de Salud Fortuny (Velutti), Distrito Sanitario Granada Metropolitano, Granada, Granada 18010, Spain

^b Consultorio de Cijuela, Distrito Sanitario Granada Metropolitano, 18339 Cijuela, Granada 18339, Spain

^c Department of Physiology, Faculty of Medicine, University of Granada, Granada, Spain

^d CIBER de Fisiopatología de la Obesidad y Nutrición (CIBEROBN), Instituto de Salud Carlos III, Granada, Spain

^e Instituto de Investigación Biosanitaria, iBs. Granada, Granada 18012, Spain

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ABSTRACT

Background and objectives: Large-scale studies (UKPDS and CALIBER) reported annual major adverse cardiovascular events (MACE) rates of 2.5–2.6% occurrence in patients with type 2 diabetes mellitus (T2D). However, data for different geographical locations in Spain are unknown. The aim of this study was to investigate the most common causes of MACE and the factors that contribute to its occurrence in patients with DM2 living in southern Spain.

Methods: In this cohort study, 297 patients with T2D were monitored over 49 months to assess the occurrence of MACE (i.e., stroke, myocardial infarction, and/or cardiovascular death). Sociodemographic and clinical data (including cardiovascular disease risk factors and initial treatments) were collected to examine their impact on the incidence of MACE.

Results: The mean age of the cohort was 67.7 years, with a male predominance (53.3%) and an average T2D duration of 8.8 years. Over the 4-year follow-up, 10.7% of patients experienced a MACE, with myocardial infarction and cardiovascular death being the most prevalent events (both 38.7%). In multivariate analysis, each additional year of T2D evolution was associated with a 15% increased risk of MACE (HR = 1.150 [1.006–1.314], $p = 0.041$), while each unit increase in estimated glomerular filtration rate (eGFR) was linked to a 3.3% risk reduction (HR = 0.967 [0.938–0.997], $p = 0.033$).

Conclusion: Myocardial infarction and cardiovascular death were the leading causes of MACE in T2D patients over a 4-year follow-up. Longer T2D duration and lower eGFR were the most significant risk factors. These findings underscore the need of closer monitoring and intensive treatment in patients with long-standing T2D and/or impaired renal function.

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Eventos cardiovasculares mayores en pacientes con diabetes mellitus tipo 2: un estudio prospectivo

RESUMEN

Introducción y objetivos: Estudios a gran escala como UKPDS y CALIBER informaron tasas anuales de eventos cardiovasculares mayores (MACE) que oscilaron entre 2,5–2,6% en pacientes con diabetes mellitus tipo 2 (DM2). Sin embargo, se desconocen las cifras relativas a distintas localizaciones geográficas españolas. El objetivo del presente trabajo fue investigar las causas más comunes de los MACE y los factores que contribuyen a su aparición en pacientes con DM2 que residen en el sur de España.

Palabras clave:

Diabetes mellitus tipo 2

Eventos cardiovasculares mayores

Enfermedad cardiovascular

Enfermedad renal crónica

* Corresponding authors.

E-mail addresses: mcgarzon@ugr.es (M.J. Castillo), amarof@ugr.es (F.J. Amaro-Gahete).

Métodos: Estudio de cohortes, 297 pacientes con DM2 fueron seguidos durante 49 meses para evaluar la aparición de MACE (accidente cerebrovascular, infarto de miocardio y/o muerte cardiovascular). Se recogieron datos sociodemográficos y clínicos para examinar su impacto en la incidencia de MACE.

Resultados: La edad media de la cohorte fue de 67,7 años, con predominio masculino (53,3%) y una duración media de la DM2 de 8,8 años. Durante los 4 años de seguimiento, el 10,7% de los pacientes experimentó MACE, siendo el infarto de miocardio y la muerte cardiovascular los eventos más prevalentes (ambos 38,7%). En el análisis multivariante, cada año adicional de evolución de la DM2 se asoció con un aumento del riesgo de MACE del 15% (HR = 1,150 [1,006-1,314], $P = 0,041$), mientras que cada unidad de aumento en la tasa de filtración glomerular estimada (eGFR) se relacionó con una reducción del riesgo del 3,3% (HR = 0,967 [0,938-0,997], $P = 0,033$).

Conclusión: El infarto de miocardio y la muerte cardiovascular fueron las principales causas de MACE en pacientes con DM2 tras 4 años de seguimiento. Una mayor duración de la DM2 y una menor eGFR fueron los factores de riesgo más significativos. Estos hallazgos subrayan la necesidad de un seguimiento más estrecho y un tratamiento intensivo en pacientes con DM2 de larga evolución y/o deterioro de la función renal.

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Introduction

Diabetes mellitus (DM), which is a high prevalent disease affecting ~537 million individuals globally (90% of whom have type 2 diabetes mellitus [T2D]), is rapidly evolving into a pandemic of considerable magnitude.¹ In Europe, there has been a noteworthy surge in the prevalence of T2D since the 1990s, with projections indicating an escalation from 6.3% in 2019 to 7.8% by 2045.^{1,2} This metabolic disorder implies several complications such as cardiovascular disease (CVD), chronic kidney disease (CKD), and other severe health disturbances. Moreover, it consistently correlates with increased rates of all-cause and CVD-specific mortality.^{3,4} Considering these public health implications, two strategies focussing of T2D management are currently considered: (i) optimizing glycemic control to alleviate associated complications, and (ii) controlling risk factors primarily linked to the onset and progression of T2D complications.^{3,5}

Given that globally the incidence of CVD in T2D is reported to be 2–3 times higher than in those without T2D, there has been an increasing emphasis on the comprehensive management of both T2D and CVD.^{1,5} Data from 2019 highlight a weighted CVD prevalence of 34.8% across 13 countries ranging from 18.0% in Saudi Arabia to 56.5% in Israel, with an overall trend of declining incidences of MACE.⁶ Recently, regulatory agencies have highlighted the importance of antidiabetic therapies to demonstrate their cardiovascular safety and efficacy with particular focus on reducing the risk of major adverse cardiovascular events (MACE).⁷ However, patients enrolled in controlled trials are typically recruited based on specific protocol criteria, a point that does not fully represent the broader patient population encountered in everyday clinical practice.⁸ Therefore, further studies are essential to improve diagnostic and therapeutic strategies aiming to mitigate the disease burden by reducing cardiovascular events.⁹

Despite significant advances in medical treatments and noticeable shifts in lifestyle habits have occurred in recent years, a significant gap remains for understanding MACE trends among patients with T2D. Furthermore, although the use of effective medications, improved medical procedures, and promising lifestyle changes programs for decreasing MACE are currently available, a lack of MACE data registered with proper quality impedes a comprehensive understanding of the practical impact of these factors.¹⁰

This study aims to elucidate the factors contributing to MACE and to identify the most common underlying causes in patients with T2D over a follow-up period of 4 years, thereby expanding the current knowledge on MACE in the south of Spain region. We hypothesize that myocardial infarction is the most frequent MACE, and that prolonged T2D duration and impaired renal function are

among the strongest predictors of cardiovascular events. Furthermore, we expect that our findings will provide novel, real-world insights into the role of traditional and emerging risk factors in T2D-associated cardiovascular disease. By doing so, this study aims to complement existing large-scale datasets and inform targeted prevention and management strategies in clinical practice.

Patients and methods

Design

This study is a prospective longitudinal cohort analysis based on a prior descriptive study (11) which primary objective was to assess the degree of glycemic control in patients with T2D.¹¹ The cohort is constructed from data sourced from two clinics in the Granada province (Spain) catering to an urban population of 18,481 individuals aged 18 years and above. Ethical considerations adhered to the Helsinki Declaration and its subsequent amendments, coupled with compliance with Spanish data protection laws. Approval was obtained from the University of Granada Ethics Committee. All participating patients were comprehensively briefed on the study protocols, both verbally and in writing, and furnished signed informed consent.

Participants

Inclusion criteria encompassed patients diagnosed with T2D within the specified health area displaying a willingness to participate in the study. Exclusion criteria included the inability to comply with follow-up requirements. The study focused on the data of T2D patients recorded in January 2017 and reevaluated in December 2020 (49 months later). A simple random sample of 297 patients was chosen from a total of 1229 T2D patients in the health area.

Outcomes

Sociodemographic variables (including gender and age) were extracted from health electronic records together with the clinical conditions. The duration of T2D was determined by reviewing the patient's history to identify when they met the specific criteria. Additionally, the body mass index (BMI) was calculated using the weight and height assessed with an electronic scale and stadiometer (SECA, model 799, Electronic Column Scale, Hamburg, Germany). Analytical parameters – including glycosylated haemoglobin (HbA1c), low density lipoprotein cholesterol (LDL-C) levels, urine albumin-creatinine ratio (uACR), and creatinine levels – were collected and analysed following routine

and validated laboratory's protocols. Controlled hypertension was defined as a systolic blood pressure (SBP) below 140 mmHg and a diastolic blood pressure (DBP) below 85 mmHg, measured in at least two consecutive clinic visits prior to baseline. Patients with SBP $\geq$ 140 mmHg or DBP $\geq$ 85 mmHg at any visit were classified as having uncontrolled hypertension. LDL-C control was defined as an LDL-C level $\leq$ 100 mg/dL, in accordance with contemporary cardiovascular risk management guidelines. HbA1c targets were assigned based on individualized patient characteristics, following the criteria outlined in the Canadian Diabetes Guidelines. The general target was HbA1c $<$ 7.0%, with a stricter goal ($\leq$ 6.5%) for younger patients without significant comorbidities, and a more relaxed target (7.1–8.5%) for older patients or those with advanced disease or high risk of hypoglycemia.¹² Medications prescribed for DM, hypertension, antiplatelet agents, anticoagulants, and lipid-lowering drugs at the start of follow-up were also recorded. Patient histories were scrutinized for emergency department visits or hospitalizations for heart failure (HF) during the follow-up period. At the end of the follow-up period, clinical histories were reviewed to identify any MACE experienced by the patients, defined as a composite of nonfatal stroke, nonfatal myocardial infarction, and cardiovascular death. MACE was defined as the occurrence of nonfatal myocardial infarction, nonfatal stroke, and cardiovascular death. The diagnosis of acute myocardial infarction was established from hospital discharge records, identified by the corresponding codes of the International Classification of Diseases, 9th and 10th Revisions (ICD-9 and ICD-10). Following the standard criteria for the definition of acute myocardial infarction, the following diagnostic elements were considered: (i) Clinical criteria, including typical symptoms of myocardial ischemia. (ii) Electrocardiographic changes compatible with acute ischemia (ST-segment elevation or depression, pathological Q waves, T-wave alterations). (iii) Cardiac biomarkers, with emphasis on the rise and/or fall of troponin above the 99th percentile of the upper reference limit. These definitions are consistent with the recommendations of international guidelines on the diagnosis of myocardial infarction.

Specific criteria and parameters were also expressed for each variable. Specifically, using the data obtained on age, sex, and creatinine levels, the estimated glomerular filtration rate (eGFR) was calculated using the CKD-EPI formula. Patients were included in the “eGFR $<$ 60 mL/min/1.73 m²” variable if they had an eGFR below this threshold in at least two separate measurements (three-month interval between them). Similarly, patients were included in the “uACR $\geq$ 30 mg/g” variable if elevated levels were observed in more than two determinations over a period exceeding three months. These thresholds for eGFR and uACR were chosen because they are the established criteria for diagnosing CKD. A standardized protocol was devised for data collection, and primary care specialist doctors were trained as volunteer evaluators.

Sample size and statistical analysis

The sample size was determined based on prior epidemiological data on MACE incidence in T2D patients. The UKPDS study reported an annual MACE incidence of approximately 2.54%, translating to an estimated 9.8% over four years. Using this as a reference, we aimed to achieve a sample size that would provide sufficient power ($\geq$ 80%) to detect clinically meaningful differences in MACE outcomes. Given a population of 1229 T2D patients in our health area, we selected a simple random sample of 297 individuals, which ensures a 95% confidence level with a margin of error below 5%. After accounting for potential dropouts ($n=6$), the final sample size remained sufficient to detect differences in key clinical variables with an estimated margin of error of 3%. This approach aligns with best practices for cohort studies investigating cardiovascular

Table 1

Descriptive characteristics of the study participants.

Sociodemographic data	
Age (y)	67.7 $\pm$ 16.0
Sex	
Men	155 (53.3%)
Women	136 (46.7%)
Clinical conditions	
Years of T2D evolution	8.8 $\pm$ 4.9
Hypertension controlled at baseline	161 (78.5%)
HbA1C	7.3 $\pm$ 1.4
Meeting LDL-C target	66 (35.1%)
eGFR $<$ 60 mL/min/1.73 m ²	78 (28.6%)
Smokers	27 (8.3%)
Patients with obesity	107 (49.3%)
HbA1c target	
$\leq$ 6.5%	13 (4.5%)
$\leq$ 7.0%	131 (45.1%)
7.1–8.0%	80 (15.5%)
7.1–8.5%	22 (27.5%)
Heart failure admission	15 (5.15%)

Data are n (%) or mean (standard deviation) unless otherwise stated. Abbreviations: eGFR, estimated glomerular filtration rate; HbA1c, Glycosylated haemoglobin; LDL-C, low-density lipoproteins cholesterol; T2D, type 2 diabetes; y, years.

outcomes and ensures adequate statistical power for multivariate analyses.¹³

All analyses were performed using SPSS (v. 22.0, IBM, Chicago, IL, USA) and R (v. 4.3.3, R Foundation, Vienna, Austria). Descriptive statistics included means $\pm$ standard deviations or medians with 95% confidence intervals for quantitative variables, and frequencies (%) for categorical variables. Data normality was assessed with the Shapiro–Wilk or D'Agostino–Pearson Omnibus test. Event-free survival graphs were generated in Python (v. 3.12.5) using pandas, numpy, and matplotlib.

For bivariate analysis, parametric or non-parametric tests were applied based on data distribution. The Chi-square test compared proportions in MACE vs. non-MACE groups, while the Fisher exact test was used if Chi-square assumptions were not met. Mean comparisons used the independent samples *t*-test or the Mann–Whitney *U* test.

Survival analysis was conducted via the Kaplan–Meier method and Log-Rank test to identify differences in event-free survival times. We then performed multivariate analyses using logistic regression, including clinically relevant and statistically significant variables from univariate testing, and verified assumptions such as residuals, heteroscedasticity, linearity, and multicollinearity (via variance inflation factors). Logistic regression was employed to identify factors associated with MACE at the end of follow-up, while Cox regression was used for time-to-event data, expressed as hazard ratios (HR). A *p*-value $<$ 0.05 was considered statistically significant.

Results

From the initial cohort of 291 patients included at the study's onset, the mean age was 67.7 years, with a slightly higher frequency in men (53.3%) than in women (46.7%). On average, participants had been diagnosed with T2D for 8.8 years, and 78.5% had controlled hypertension at baseline. Regarding glycemic control, the average HbA1C levels were 7.3. A relatively low percentage of participants (35.1%) met the LDL-C target at baseline, and 28.6% had estimated glomerular filtration rate (eGFR) $<$ 60 mL/min/1.73 m². A low proportion of participants were smokers (8.3%) and nearly half of them (49.3%) were categorized as individuals with obesity. Finally, only 5.15% had admissions due to HF (Table 1).

Table 2

Parameters associated with the occurrence of major adverse cardiovascular events.

	MACE				
Subgroup	Yes (N = 31)	No (N = 260)		HR (95% CI)	P-value
Sociodemographic data					
Age (y)	70.13±10.22	67.37±10.92		1.03 (0.99, 1.06)	0.100
Sex					
Women	14(45.2%)	122(46.9%)		0.86 (0.42, 1.78)	0.700
Men	17(54.8%)	138(53.1%)		0.86 (0.42, 1.78)	0.700
Clinical conditions					
Years of T2D evolution	10.97±4.69	8.58±4.86		1.09 (1.02, 1.17)	0.020
Uncontrolled hypertension	17 (65.4%)	144(80.4%)		0.56 (0.24, 1.29)	0.200
HbA1c mean	7.24±1.25	7.31±1.44		0.96 (0.73, 1.27)	0.800
Meeting LDL-C target	9(45%)	113(67.3%)		2.64 (1.06, 6.56)	0.036
eGFR	59.69±23.73	74.72±21.14		0.97 (0.96, 0.99)	0.001
eGFR <60mL/min/1.73m2	13(50%)	65(26.3%)		2.44 (1.11, 5.34)	0.026
uACR	16.00 ± 27.18	4.88±13.27		1.02 (1.01, 1.04)	<0.001
uACR ≥ 30 mg/g	7(28%)	13(5.4%)		6.21 (2.57, 15.00)	<0.001
BMI	29.11±3.23	30.64±5.27		0.94 (0.86, 1.04)	0.200
Heart failure admission	8(25.8%)	7(2.7%)		9.35 (4.15, 21.06)	<0.001
Treatment					
Metformin	24(77.4%)	216(83.1%)		0.71 (0.30, 1.65)	0.420
Sulfonylureas	8(25.8%)	61(23.5%)		1.17 (0.52, 2.62)	0.710
DPP4 inhibitors	6(19.4%)	56(21.5%)		0.91 (0.37, 2.23)	0.840
SGLT2 inhibitors	4(12.9%)	13(5%)		1.94 (0.59, 6.40)	0.276
GLP1 analogs	0(0%)	13(5%)			0.990
Insulin	10(32.3%)	68(26.2%)		1.19 (0.54, 2.60)	0.700
Acetylsalicylic acid	9(29%)	75(28.8%)		0.92 (0.41, 2.07)	0.847
Anticoagulants	8(25.8%)	18(6.9%)		3.32 (1.43, 7.67)	0.005
Diuretic	21(67.7%)	116(44.8%)		2.36 (1.10, 5.04)	0.027
Statins	15(48.4%)	110(42.3%)		1.18 (0.57, 2.41)	0.655
Ezetimibe	3(9.7%)	12(4.6%)		1.42 (0.34, 5.96)	0.632

Data are n (% by column) or mean (standard deviation). The HR are expressed with their 95% confidence interval (IC). Abbreviations: BMI, body mass index; DPP4 inhibitors, dipeptidyl peptidase 4 inhibitors; eGFR, estimated glomerular filtration rate; GLP1 analogs, GLP-1 peptide analogs; HbA1c, glycosylated haemoglobin; HR, hazard ratios; LDL-C, low-density lipoproteins cholesterol; MACE, major adverse cardiovascular events SGLT2 inhibitors, Sodium-Glucose cotransporter type 2 inhibitors; T2D, type 2 diabetes; uACR, urine albumin-creatinine ratio; y, years. *p* value of analysis of variance analysis between groups.

Data from the 4-year follow-up period are presented in Table 2. A total of 10.7% (*n* = 31) of the patients experienced a MACE. Among these, myocardial infarction accounted for 38.7% of the events, and cardiovascular death also accounted for 38.7%. Stroke was the third most common cause of MACE representing 29.0%. Additionally, 6.5% of patients experienced more than one type of MACE during the follow-up period. The annual MACE rate was 2.61%, with the annual rate of nonfatal myocardial infarction being 1.03%, cardiovascular death at 1.01%, and nonfatal stroke at 0.76%.

Moreover, a subgroup analysis by sex and age did not reveal statistically significant differences in MACE risk (sex: HR = 0.865 [0.42–1.78], *p* = 0.70; age: HR = 1.027 [0.993–1.062], *p* = 0.10; Table 2). Among sociodemographic variables, the duration of T2D was significantly associated with MACE with individuals who had a longer history of T2D showing a higher risk (HR = 1.092 [1.016–1.173]; *p* = 0.020; Table 2 and Fig. 1). Both eGFR < 60 mL/min/1.73 m² and a decline in eGFR were also positively associated with MACE (HR = 2.438 [1.113–5.345]; *p* = 0.026 and HR = 0.972 [0.956–0.989]; *p* = 0.001, respectively; Table 2 and Fig. 1). Moreover, elevated urine albumin-creatinine ratio (uACR) levels and uACR ≥ 30 mg/g were significantly correlated with a

higher risk of MACE (HR = 1.024 [1.010–1.037]; *p* < 0.001 and HR = 6.213 [2.573–15.000]; *p* < 0.001, respectively; Table 2 and Fig. 1). Achieving the LDL-C target at baseline was related to lower MACE risk (HR = 2.640 [1.062–6.563]; *p* = 0.036; Table 2 and Fig. 1).

Several medications were found to be significantly associated with the risk of MACE. The use of anticoagulants and diuretics was linked to a substantially higher MACE risk (HR = 3.317 [1.434–7.671]; *p* = 0.005 and HR = 2.360 [1.104–5.041]; *p* = 0.027, respectively; Table 2 and Fig. 1). Other factors such as gender, age, BMI, and other medications (including all antidiabetic drugs) did not show a significant association with MACE.

The multivariate analysis indicated that years of T2D evolution (HR = 1.150 [1.006–1.314]; *p* = 0.041) and eGFR (HR = 0.967 [0.938–0.997]; *p* = 0.033) were significant predictors of MACE (Table 3).

Discussion

The present study aimed to identify the primary causes of MACE in Spanish individuals with T2D over a four-year period.

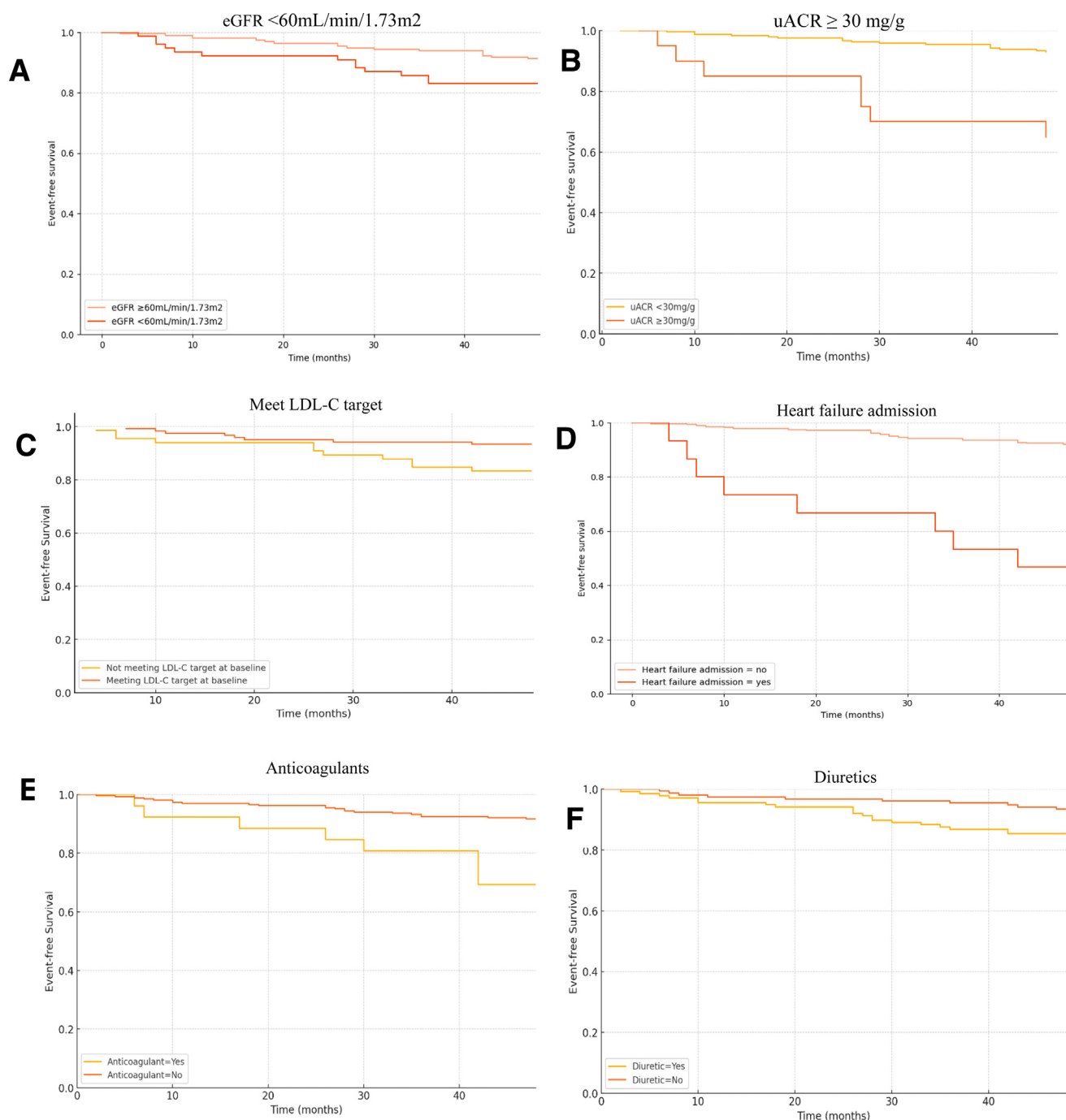


Fig. 1. Kaplan–Meier survival curves for major adverse cardiovascular events of patients with type 2 diabetes depending on the clinical features (Panels A–D) or drugs studied (Panels E, F) that obtained statistical significance. Panel (A) represents the presence of $\text{eGFR} < 60 \text{ mL/min/1.73 m}^2$, and Panel (B) represents $\text{uACR} \geq 30 \text{ mg/g}$, Panel (C) patients who meet LDL-C target at baseline of study, Panel (D) represent heart failure admission, Panel (E) represents anticoagulants and Panel (F) represents diuretics. Abbreviations: eGFR, estimated glomerular filtration rate; LDL-C, low-density lipoproteins cholesterol, uACR, urine albumin-creatinine ratio.

Overall, myocardial infarction and cardiovascular death were the most common causes of MACE in our study cohort. Moreover, a longer duration of T2D, failure to meet LDL-C targets, lower eGFR, increased uACR, and the use of diuretics and anticoagulants were identified as significant risk factors for MACE.

Our findings can be compared to those of the United Kingdom Prospective Diabetes Study (UKPDS), which reported a MACE rate of 2.54% (2.61% in our study), a nonfatal myocardial infarction rate of 0.90% (1.01% in our study), a cardiovascular death rate of 1.12% (1.01% in our study), and a nonfatal stroke rate of 0.52% (0.76% in our study).¹³ It should be noted that the UKPDS had strict inclusion

criteria, an issue that may potentially explain the slightly lower rates observed in some specific outcomes.¹³

The CALIBER programme – a database comprising 1.9 million people in the United Kingdom – reported a MACE rate of 2.64%, a nonfatal myocardial infarction rate of 1.13%, a cardiovascular death rate of 0.75%, and a nonfatal stroke rate of 0.75%,¹⁴ data that are quite similar than those obtained in the present work. These suggest that our cohort study accurately reflects the broader population trends observed in large-scale studies.

In contrast, the CICCOR registry – which focuses on patients with pre-existing chronic coronary syndrome in the adjacent region

Table 3
Multivariate analysis of variables related to major adverse cardiovascular events.

	HR (95% CI)	p-Value
Years of T2D evolution	1.150 (1.006, 1.314)	0.041
eGFR	0.967 (0.938, 0.997)	0.033
uACR	1.014 (0.982, 1.046)	0.389
Meeting LDL-C target	1.048 (0.276, 3.978)	0.945
Anticoagulant treatment	4.157 (0.797, 21.688)	0.091
Diuretic treatment	2.574 (0.589, 11.251)	0.209

The hazard ratios are expressed with their 95% confidence interval (CI). Abbreviations: eGFR, estimated glomerular filtration rate; HR, hazard ratio; LDL-C, low-density lipoproteins cholesterol; T2D, type 2 diabetes mellitus; uACR, urine albumin-creatinine ratio. *p* value of analysis of multivariate analysis between groups.

of Córdoba (Spain) – exhibited significantly higher rates compared with our data reporting a MACE rate of 6.5%, a nonfatal myocardial infarction rate of 1.97%, a cardiovascular death rate of 4.69%, and a nonfatal stroke rate of 1.64%.¹⁵ These increased rates may be explained due to the higher-risk profile of this patient subpopulation.^{15,16}

Pharmacological clinical studies such as EMPAREG and DECLARE-TIMI reported MACE rates ranging from 2.26% to 4.39%, while the recent SURPASS-CVOT study informed a MACE rate below 3.5%.^{17–19} These findings place our study's results within the lower to mid-range of these pivotal trials, underscoring the relevance and reliability of our real-world data at reflecting contemporary cardiovascular risk trends. Our findings on MACE incidence in T2D patients align with those reported in major cardiovascular outcome trials, yet certain discrepancies highlight key differences in study design and patient selection criteria. The EMPAREG trial, which evaluated empagliflozin in high-risk patients, reported a MACE rate of 2.26% per year, slightly lower than our study (2.61%), likely reflecting the cardioprotective effects of SGLT2 inhibitors, which were not universally used in our cohort.¹⁷ Similarly, DECLARE-TIMI—a large-scale trial assessing dapagliflozin—reported an annual MACE incidence of 2.4%, in line with our findings.¹⁸ These results reinforce the notion that contemporary real-world MACE rates in T2D remain relatively stable when considering patient populations receiving standard care. In contrast, the SURPASS-CVOT trial observed a MACE incidence above 3.5%, significantly higher than in our study.¹⁹ This discrepancy is likely attributed to differences in patient selection, as SURPASS-CVOT focused exclusively on individuals with established atherosclerotic cardiovascular, thereby representing a secondary prevention cohort with inherently higher baseline risk. Conversely, our study includes a broader T2D population, encompassing both primary and secondary prevention settings, which may account for the lower MACE rates observed. These comparisons emphasize that while our study aligns with the general trends observed in large cardiovascular outcome trials, differences in patient characteristics, cardiovascular disease burden, and therapeutic interventions should be carefully considered when interpreting MACE incidence across different cohorts.

The duration of T2D evolution was significantly associated with the occurrence of MACE, with each additional year of T2D increasing the risk of experiencing a cardiovascular event by 15%. This result aligns with a study conducted in South Korea, where longer T2D duration resulted in an 8.6% MACE incidence, compared to 5.9% in those with shorter duration.²⁰ These findings are consistent with the Framingham Heart Study, which identified that every 10-year increase in diabetes duration was associated with a 38% increased risk of coronary heart disease, independent of other cardiovascular risk factors.²¹ Notably, it is striking that the duration of T2D evolution holds significance above the individual's age. This point emphasizes the importance of the cumulative vascular damage due

to prolonged exposure to chronic hyperglycemia, independent of the patient's age, a finding that is supported by a meta-analysis of studies from 30 countries, which found that younger rather than older age at T2D diagnosis was associated with a higher risk of vascular disease.²²

We also identified eGFR as a significant factor since for each unit increase in eGFR, a 3.3% reduction in the risk of MACE was obtained. This finding highlights the importance of maintaining a proper renal function to reduce cardiovascular risk in patients with T2D. A previous study has shown that renal function can reflect microvascular damage, a protective factor against cardiovascular events.²³ Consistently, a decline in eGFR has been associated with an increased risk of CVD events.²⁴ Indeed, in patients with T2D, a steeper decline in the eGFR slope is linked to a higher risk of cardiovascular events, even after adjusting for traditional cardiovascular risk factors including albuminuria.^{23,25}

Meeting LDL-C targets showed significance in the univariate but not in the multivariate analysis, suggesting that other factors may have influenced the results. Although the HR indicated a slight risk increase, the wide confidence interval and lack of significance advocate that LDL-C target achievement did not clearly reduce the risk in our study. However, it is well established that meeting LDL-C goals is crucial for lowering cardiovascular events in T2D patients.^{26,27} Regarding glycemic control, both our study and other studies found no association between HbA1c levels and MACE.²⁸ Nevertheless, we cannot rule out the influence of blood glucose levels, as other research pointed out that glucose time in range may be a more effective predictor of reduced MACE risk.²⁸

Although anticoagulant treatment did not reach statistical significance in the multivariate analysis, a trend towards a higher risk of MACE in patients undergoing anticoagulant therapy was noted. This finding could reflect a greater burden of comorbidities or more severe disease in these individuals, rather than a direct effect of the anticoagulant treatment itself. Similarly, despite the common use of diuretics in managing hypertension and HF in patients with T2D, their association with MACE may be influenced by other comorbidities and longer disease duration. Diuretics, particularly loop diuretics, are associated with increased cardiovascular risk, potentially due to underlying conditions.²⁹ Thiazides, while beneficial for blood pressure control, may increase MACE in certain populations possibly due to risks of hypoperfusion and arrhythmias, particularly in patients with intensive BP control.³⁰ Therefore, the use of these medications should be tailored to individual patient profiles, balancing the benefits and risks to optimize cardiovascular outcomes.

Given that prolonged diabetes duration and declining renal function were the strongest predictors of MACE, we emphasize the importance of routine renal function monitoring (eGFR and uACR) and early nephroprotective interventions such as SGLT2 inhibitors. Additionally, achieving LDL-C targets (<70 mg/dL for high-risk and <55 mg/dL for very high-risk patients) is crucial to reducing cardiovascular events, reinforcing the need for statin therapy intensification when necessary. From a healthcare system perspective, integrated care models combining primary care, endocrinology, nephrology, and cardiology can enhance risk stratification, while population-based screening programs may facilitate early intervention in high-risk patients. Our findings highlight the need for a proactive, multifactorial approach that prioritizes renal function assessment, lipid control, and coordinated cardiovascular management to improve long-term outcomes in T2D patients.

While the results of our study are of scientific and clinical interest, it is important to acknowledge some limitations. Firstly, the small sample size may have restricted our ability to detect statistically significant outcomes. Larger-scale studies are warranted to confirm our findings. Additionally, the generalizability of our results may be limited by regional differences, such as the

unique characteristics of the T2D population in the south of Spain compared to those in other parts of the world. This highlights the necessity for more comprehensive data to accurately determine trends across different geographical regions and ethnicities worldwide. Lastly, we did not collect information on genetic predispositions or lifestyle factors, which could also influence MACE occurrence and thus represent another important limitation of our study.

Conclusions

Our study reveals that myocardial infarction and cardiovascular death are the main causes of MACE in a Spanish population with T2D over a 4-year follow-up. Importantly, longer diabetes duration and declining renal function emerged as the strongest predictors of MACE risk, with each additional year of T2D increasing cardiovascular risk by 15% and each unit increase in eGFR reducing risk by 3.3%. These findings underscore the need for proactive cardiovascular risk management, particularly in patients with long-standing T2D or chronic kidney disease. Routine monitoring of renal function (eGFR and uACR) should be integrated into standard clinical practice to facilitate early identification of high-risk individuals. Additionally, achieving LDL-C targets remains a key priority in cardiovascular prevention. To improve long-term outcomes, healthcare systems should implement multifactorial interventions that include personalized treatment strategies, statin therapy intensification, and early introduction of nephroprotective agents (such as SGLT2 inhibitors). Future research should explore how integrating genetic, lifestyle, and socioeconomic factors into cardiovascular risk models could further enhance patient stratification and optimize therapeutic approaches in T2D management.

Ethical considerations

Ethical considerations adhered to the Helsinki Declaration and its subsequent amendments, coupled with compliance with Spanish data protection laws. Approval was obtained from the University of Granada Ethics Committee. All participating patients were comprehensively briefed on the study protocols, both verbally and in writing, and furnished signed informed consent.

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Conflict of interest

The authors declare that they have no conflict of interest concerning the research, authorship, and/or publication of this article.

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